2021 Ph H2 Q9

Section: Electricity

Topic: Forces on Teeth, Irradiance, Semiconductors

Dental braces and LED curing lamp. (a)(i) Calculate resultant force on tooth. (a)(ii) Explain why tooth does not move sideways. (b)(i) Find minimum exposure time for cement to harden. (b)(ii) Show LED behaves as point source using inverse square law. (b)(iii) (A) Define doped semiconductor. (B) Explain LED emission by band theory.

Worked solution

(a)(i) Two equal forces $F = 19.5$ N act symmetrically at 14° to backward direction.
Resultant = $2F \cos(14°)$
= $2 \times 19.5 \times \cos(14°) = 37.8$ N

Answer: 37.8 N

(a)(ii) Horizontal components of the two equal forces cancel, so no net sideways force acts on the tooth.

(b)(i) Power incident = $IA = 11800 \times 1.24 \times 10^{-5} = 0.146$ W
Time = $E/P = 2.10/0.146 = 14.4$ s

Answer: 14.4 s

(b)(ii) For point source: $I \propto 1/d^2 \Rightarrow I \cdot d^2$ constant.
Values: 0.57, 0.56, 0.57, 0.58 W m
≈ constant (avg ≈ 0.57).

Therefore LED acts as a point source over this range.

(b)(iii)(A) A doped semiconductor is one in which small amounts of impurities (donors or acceptors) are added to increase the number of free charge carriers.

(b)(iii)(B) In forward bias, electrons from the n-side recombine with holes on the p-side. When electrons drop from conduction band to valence band across the band gap, they release energy as photons of light.

Final answers

(a)(i) Resultant force ≈ 37.8 N

(a)(ii) Horizontal forces cancel

(b)(i) $t \approx 14.4$ s

(b)(ii) $I \cdot d^2 \approx$ constant ⇒ point source

(b)(iii)(A) Doped semiconductor: impurities increase carriers

(b)(iii)(B) Light from e^-–hole recombination across band gap

Revision tips

- Forces: resolve into components, symmetry cancels sideways parts.
- Irradiance $I = P/A$, energy = power × time.
- Inverse square law: $I \propto 1/d^2$ for point sources.
- Semiconductors: doping increases n- or p-type carriers.
- LED emission = recombination of electrons and holes releasing photons.